

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2023/2024

SEPTEMBER EXAMINATION

UCCD1024 DATA STRUCTURE AND ALGORITHMIC PROBLEM SOLVING

THURSDAY, 5 OCTOBER 2023

TIME : 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMPUTER ENGINEERING

This question paper consists of **THREE (3)** questions in Section A and **TWO (2)** questions in Section B.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**. Each question carries 25 marks.

Should a candidate answer more than ONE (1) question in Section B, marks will only be awarded for the **FIRST** questions in the order the candidate submits the answers.

Candidates are allowed to use a non-scientific calculator.

Answer questions only in the answer booklet provided.

UCCD1024 DATA STRUCTURE AND ALGORITHMIC PROBLEM SOLVING**SECTION A**

- Q1. (a) Given the segment of codes, answer the following questions based on the segment of codes. State whether the `cout` statement is causing error: Provide the output of the `cout` statements if no error. Provide reason if the `cout` statement causes error.

<pre> struct pet{ char type[20]; char breed[20]; int age; char gender; char colour[10]; pet *next; //type, breed, age, gender, // colour pet(const char[], const char[],int,char,const char[]); }; </pre>	<pre> int main(){ pet cat ("Feline", "Siamese", 3, 'F', "Silver"); pet dog("Canine", "Poodle", 5, 'M', "Brown"); pet *ptr; ptr = &cat; cat.next = &dog; dog.next = NULL; return 0; } </pre>
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- (i) `cout << (*ptr).breed[5] << "\n";` (2 marks)
- (ii) `cout << ptr->colour << "\n";` (2 marks)
- (iii) `cout << ptr->next->type[1] << "\n";` (2 marks)
- (iv) `cout << cat.next.gender << "\n";` (2 marks)

- (b) Answer the following questions based on the segment of codes below.

<pre> struct Node { char item; Node *next; Node(char newItem){ item = newItem; next = NULL; } }; </pre>	<pre> int main(){ Node item1('B'); Node item2('D'); Node item3('A'); Node item4('E'); Node item5('C'); Node *head; head = &item1; item1.next = &item2; item3.next = head; head = &item3; return 0; } </pre>
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UCCD1024 DATA STRUCTURE AND ALGORITHMIC PROBLEM SOLVING**Q1. (b)(Continued)**

- (i) Draw the linked list that is created after the segment of codes in `main()` (4 marks)
- (ii) Write statements to insert node with value 'C' as the third node in the linked list created in `main()`. (4 marks)
- (iii) Write statement to insert node with value 'E' as the last node in the linked list after you inserted 'C' in Q1(b)(ii). (2 marks)
- (iv) Write statement to remove node with value 'D' in the linked list after you have inserted 'E' in Q1(b)(iii). (2 marks)
- (v) Write a function `void print(Node *head)` to print the content in a linked list. The head pointer is pointing to the first node in the list when pass to function. (5 marks)

[Total : 25 marks]

Q2. (a) The following questions are related to Stack and Queue.

- (i) Briefly describe the difference between Stack and Queue in terms of item insertion and deletion. (3 marks)
- (ii) The following pseudo-code is executed on a queue and a stack.

```

Create a queue
Create a stack

while( not end of file )
    read number

    if( number not 0 )
        push the number onto the stack
    else
        pop off one number to x

        while( stack is not empty)
            pop off one number to x
            enqueue x + 8 to the queue
        end while
    end if
end while

```

Given the following data: 5, 7, 12, 4, 0, 4, 6, 8, 67, 34, 5, 0, 44, 33, 22, 6, 0, show the contents of the queue after each 0 is inserted.

(6 marks)

UCCD1024 DATA STRUCTURE AND ALGORITHMIC PROBLEM SOLVING**Q2. (a)(Continued)**

- (iii) s1 and s2 are two empty stacks of integers. Draw the content of s1 and s2 before and after the loop operations as shown below.

```
pushStack(s1, 5)
pushStack(s1, 9)
pushStack(s1, 2)
pushStack(s1, 3)
pushStack(s1, 1)
pushStack(s1, 8)

loop not emptyStack(s1)
  x = popStack(s1)
  if x%2 equal 0
    pushStack(s2, x)
  else
    pushStack(s2, x+1)
end loop
```

(6 marks)

- (b) Given the following data structures,

Linked List, Tree, Binary Search Tree, Double Linked List, Stacks Queues, Circular Linked List,

suggest an appropriate data structure to solve the following problems and justify your answer.

- (i) You want to keep track of appointment and event according to time and date. It is important to find all the events between starting date/time and ending date/time. (2 marks)
- (ii) You want to build a snake and ladder board game. When the player encounter a ladder then it will lead player up to a higher tile value. However, when the player encounter a snake then it will lead player to a lower tile value. (2 marks)
- (iii) You want to build a file explorer where it will allow you to view file in different folders in the computer. Every folder can be nested in many levels. (2 marks)
- (iv) Provide the big-O notation for best and worst cases when performing searching using Binary Search and Hash Table. (4 marks)

[Total : 25 marks]

UCCD1024 DATA STRUCTURE AND ALGORITHMIC PROBLEM SOLVING

Q3. (a) Given an array: `int arr[8]={9, 3, 4, 7, 8, 1, 5, 2}`, draw a diagram to show how array `arr` is sorted in ascending order using Mergesort algorithm. (8 marks)

(b) The following fruit names are inserted to the hash table from left to right. Table 1.0 shows the hash value for each fruit name in a hash table. Draw a hash table based on this Table 1.0. Use chaining to solve collision when it occurs. Show the NULL value in the hash table.

Mango, Banana, Orange, Apple, Grape, Durian, Jackfruit, Mangosteen

Name	Hash Value
Mango	1
Banana	3
Orange	7
Apple	2
Grape	7
Durian	0
Jackfruit	0
Mangosteen	1

Table 1.0 : Hash Table Value

(6 marks)

(c) The function below is used to print all the nodes in a linked list. Derive the Big-O notation to print a list using the function below. Explain the steps on how to calculate the time complexity for the function and derivation of the Big-O notation.

Assume that the `size()` do not require any step and the `get()` will require n steps where n is the size of the linked list. The `cout` and `return` statements also do not require any step to perform.

```
bool printList(List list){ //assume list contains n nodes
    type temp;
    int n;
    n = list.size(); //size() do not require any step when
                    //called

    for(int i=1; i<=n; i++) {
        if(!list.get(i, temp)) //get() required n steps
                                //when called
            return false;
        cout << temp << " ";
    }
    return true;
}
```

(7 marks)

UCCD1024 DATA STRUCTURE AND ALGORITHMIC PROBLEM SOLVING**Q3. (Continued)**

- (d) Given the following data structure,

Linked List, Tree, Binary Search Tree, Double Linked List, Stacks, Queues, Circular Linked List, Graph, Hash Table,

suggest an appropriate data structure to solve the following problems and justify your answer.

- (i) You want to store the license plates for all the cars in Malaysia and must be able to retrieve the license plate in the fastest time possible. (2 marks)
- (ii) You want to build a text editor where you can go forward and backward to any words in the text. The text editor also able to insert or delete any word from anywhere in the text. (2 marks)

[Total : 25 marks]

SECTION B

- Q4. (a) (i) Figure 1.0 below shown the Binary Search Tree. Redraw the Binary Search Tree after the following integers are inserted into the existing Binary Search Tree from left to right one at a time.

Integers: 12 26 37 17 51

(5 marks)

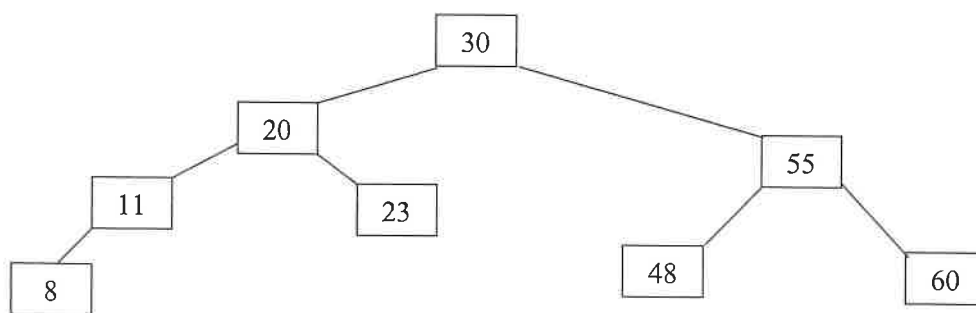


Figure 1.0 Binary Search Tree

UCCD1024 DATA STRUCTURE AND ALGORITHMIC PROBLEM SOLVING**Q4. (a)(Continued)**

- (ii) Draw the resulting binary search tree after the following remove nodes statements are executed (after inserting the 5 nodes) that you drawn in Q4(a)(i). (5 marks)

```
tree.remove(20);
tree.remove(26);
tree.remove(55);
```

- (iii) Perform balancing check on the binary search tree (after removing nodes 20, 26, and 55) that you draw in Q4(a)(ii). If a node is not balanced, you need to balance it first.
Draw the final tree after you have performed balancing check and balanced the unbalance node for all the nodes in the tree. (5 marks)

- (b) Write a function `void BST::printLevelNodes2(BTNode *cur, int cnt, int level)` to traverse the binary search tree recursively and print nodes in each level of the tree. Refer to example output below when the function `printLevelNodes` is called to print nodes in each level for Binary Search Tree shown in Figure 1.0.

Sample Output:

Level 1 nodes:

30

Level 2 nodes:

20 55

Level 3 nodes:

11 23 48 60

Level 4 nodes:

8 12 26 37 51

Level 5 nodes:

17

(6 marks)

- (c) Assume that all the nodes in a binary search tree have integers value between 1 and 1000 and we want to search for the node with value 100. State whether the following sequences could be the sequence of nodes examined in order to find 100. Justify your answer if it is not possible.

- (i) 55, 55, 68, 69, 90, 99 (2 marks)

- (ii) 600, 23, 89, 73, 78, 83 (2 marks)

[Total : 25 marks]

UCCD1024 DATA STRUCTURE AND ALGORITHMIC PROBLEM SOLVING

- Q5. (a) Answer the following questions using the Undirected Graph shown in figure 2.0.

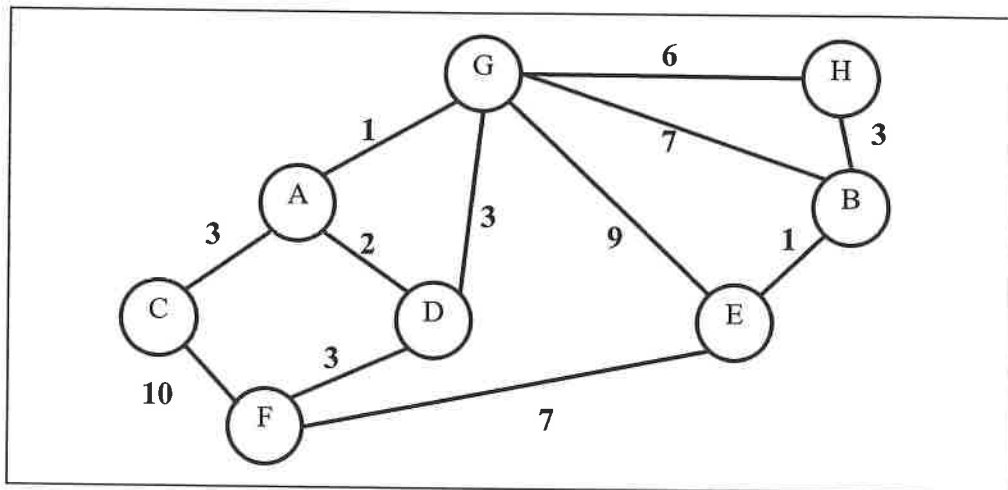


Figure 2.0 Undirected Graph

- (i) Construct an adjacency list representation from the Undirected Graph. (4 marks)
 - (ii) Perform Depth-First Search on the graph and list out the order of the nodes visited. Start from node A. Visit the neighboring nodes according to alphabetical order. (4 marks)
 - (iii) Perform Breadth-First Search on the graph and list out the order of the nodes visited. Start from node A. Visit the neighboring nodes according to alphabetical order. (4 marks)
 - (iv) Find the Shortest Path in Directed Graph A using Dijkstra algorithms from vertex A. Show the final graph after all the shortest path to all vertices from vertex A are found. (5 marks)
- (b) Describe what are the base case and the recursive case for the following problems if each question will be solved using recursion.
- (i) To search for a word in a dictionary using Binary Search. (2 marks)
 - (ii) To find the maximum integer in an array. (2 marks)
- (c) Write a recursive function `int addArray(int arr[], int n)` to add all the elements value in array arr. The n variable is the size of the array. (4 marks)
- [Total : 25 marks]